

**UNIT II**  
**INHERITANCE AND POLYMORPHISM**

**Course Objective - Cob1 :** To study the properties and usage of inheritance and polymorphism in C++.

**Course Outcome - CO1 :** Develop a C++ program using inheritance and polymorphism concept.

**Bloom's Taxonomy Level: K1, K2, K3, K4, K5, K6**

**Part A**

| S. No. | Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | BTL* |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 1.     | <b>Define Inheritance?</b> <ul style="list-style-type: none"> <li>Inheritance is the process by which objects of one class acquire the properties of objects of another class.</li> <li>It provides the idea of reusability.</li> <li>This is possible by deriving a new class from the existing class . The new class will have the combined features of both the classes.</li> </ul>                                                                                                                                                                                                          | K1   |
| 2.     | <b>What is meant by subclass and superclass.</b><br><b>Subclass:</b> a class which has link to a more general class<br><b>Superclass:</b> a class which has one or more members which are classes themselves.                                                                                                                                                                                                                                                                                                                                                                                   | K1   |
| 3.     | <b>What are the types of inheritance?</b> <p>(i).<b>Single inheritance:</b> A derived class with only one base class is called single inheritance</p> <p>(ii).<b>Multiple inheritance:</b> A class can inherit properties from more than one class which is known as multiple inheritance</p> <p>(ii).<b>Multilevel inheritance:</b> A class can be derived from another derived class which is known as multilevel inheritance.</p> <p>(iv).<b>Hierarchical inheritance:</b> When the properties of one class are inherited by more than one class, it is called hierarchical inheritance.</p> | K1   |
| 4.     | <b>Show how to define derived classes?</b><br>A derived class can be defined by specifying its relational ship with the base class in addition to its own details.<br><b>The syntax is:</b><br><pre>class derived-class-name : visibility-mode base-class-name { ..... ..... };</pre> Here the visibility mode is optional and if present, may be either private or public. The default mode is private.                                                                                                                                                                                        | K2   |
| 5.     | <b>Define abstract class.</b><br>Abstract class is one that is not used to create objects. An abstract class is designed only to act as a base class. It is a design concept in program development and provides a base upon which other classes may be built.                                                                                                                                                                                                                                                                                                                                  | K1   |
| 6.     | <b>Define Polymorphism?</b> <ul style="list-style-type: none"> <li>It means the ability to take more than one form.</li> <li>For ex, consider the operation of addition.</li> <li>If the operands are strings, then the operation would produce a third string by concatenation.</li> <li>If the operands are numbers, it will generate a sum.</li> </ul>                                                                                                                                                                                                                                       | K1   |



|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |    |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 7.  | <b>What are the types of Polymorphism?</b><br><p>Compile time polymorphism (Static binding)</p> <p>Run time polymorphism (Dynamic binding)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | K1 |
| 8.  | <b>Define Static Binding.</b><br><p>Static binding means the code associated with a given procedure call is known at the compile time.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | K1 |
| 9.  | <b>Define dynamic binding.</b> <ul style="list-style-type: none"> <li>It refers to the linking of a procedure call to the code to be executed in response to the call.</li> <li>Dynamic binding means that the code associated with a given procedure call is not known until the time of the call at run time.</li> <li>It is associated with polymorphism and inheritance</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                 | K1 |
| 10. | <b>Define function overloading.</b><br><p>More than one functions with same name but different arguments is known as function overloading.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | K1 |
| 11. | <b>What is operator overloading?</b><br><p>Operator overloading is a compile-time polymorphism in which the operator is overloaded to provide the special meaning to the user-defined data type. Operator overloading is used to overload or redefines most of the operators available in C++.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | K1 |
| 12. | <b>List the Operator that cannot be overloaded.</b><br><p>Operator that cannot be overloaded are as follows:</p> <ul style="list-style-type: none"> <li>Scope resolution operator (::)</li> <li>Size operator (sizeof)</li> <li>Class member access operator (.)</li> <li>Pointer to member operator (.*)</li> <li>Ternary (or) Conditional operator(?:)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                    | K1 |
| 13. | <b>Explain the process of operator overloading?</b> <ol style="list-style-type: none"> <li>Create a class that defines the data type that is to be used in the overloading operation.</li> <li>Declare the operator function operator op() in the public part of the class. It may be either a member function or a friend function.</li> <li>Define the operator function to implement the required operations.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                            | K2 |
| 14. | <b>Outline the rules for overloading operators?</b> <ul style="list-style-type: none"> <li>Existing operators can only be overloaded, but the new operators cannot be overloaded.</li> <li>The overloaded operator contains atleast one operand of the user-defined data type.</li> <li>We cannot use friend function to overload certain operators. However, the member function can be used to overload those operators.</li> <li>When unary operators are overloaded through a member function take no explicit arguments, but, if they are overloaded by a friend function, takes one argument.</li> <li>When binary operators are overloaded through a member function takes one explicit argument, and if they are overloaded through a friend function takes two explicit arguments.</li> </ul> | K2 |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |    |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 15. | <p><b>What is the purpose of using operator function? Write its syntax.</b></p> <p>To define an additional task to an operator, we must specify what it means in relation to the class to which the operator is applied. This is done by Operator function , which describes the task. Operator functions are either member functions or friend functions. The general form is</p> <pre>return type classname :: operator op(arglist ) { function body }</pre> <p>where return type is the type of value returned by specified operation.<br/>Op-operator being overloaded. The op is preceded by a keyword operator. operator op is</p> | K1 |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
|     | the function name.                                                                                                                                                                                                                                                                                                                                                                                                                                 |    |
| 16. | <b>Explain how will you overload Unary &amp; Binary operator using member functions?</b> <ul style="list-style-type: none"> <li>When unary operators are overloaded using member functions it takes no explicit arguments and return no explicit values.</li> <li>When binary operators are overloaded using member functions, it takes one explicit argument. Also the left hand side operand must be an object of the relevant class.</li> </ul> | K2 |

### **Part B**

| S.No. | Question                                                                                                                                                       | BTL* |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 1.    | Explain the various types of Inheritance with suitable example program.                                                                                        | K2   |
| 2.    | Develop a c++ program to implement Single Inheritance.                                                                                                         | K3   |
| 3.    | Develop a c++ program to implement Multi level Inheritance.                                                                                                    | K3   |
| 4.    | Identify the suitable type of inheritance method and develop a code to Illustrate the scenario where a class can be inherited from more than one base classes. | K3   |
| 5.    | What are abstract classes? Give an example program to illustrate the use of abstract classes.                                                                  | K2   |
| 6.    | Develop a c++ program to implement function overloading concept using constructor functions.                                                                   | K3   |
| 7.    | Elaborate on Unary operator overloading with an example.                                                                                                       | K6   |
| 8.    | Elaborate on binary operator overloading with an example.                                                                                                      | K6   |